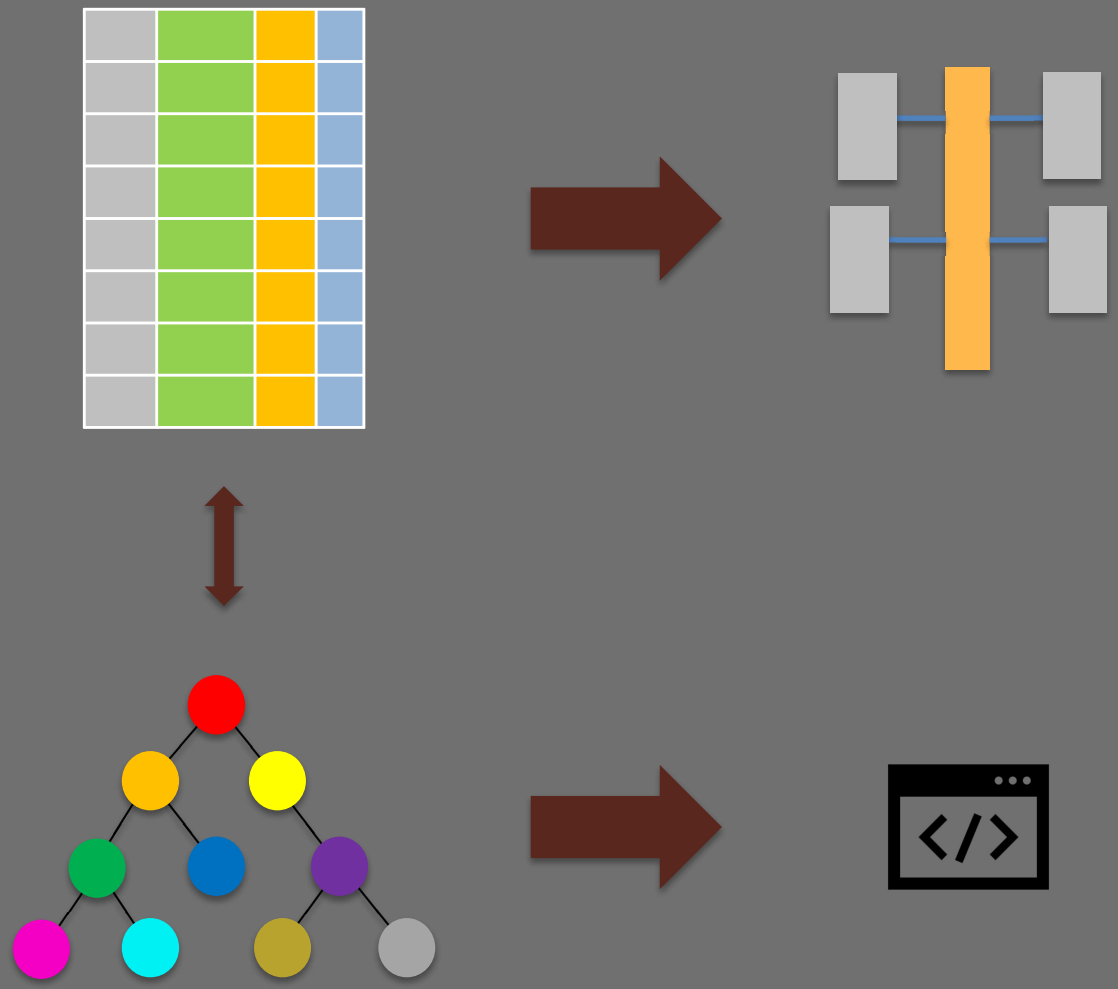
Design Review



Multiple databases

Each specialized to its role.

OLTP

- Transactional → normalized for data integrity
- Entity-relationship design

OLAP

Analytical → denormalized for efficiency

Dimensional modeling

- Facts: Data to be aggregated
- Dimensions: Data to aggregate over
- Replicate, pre-compute data for efficiency.
- From ERD, try collapsing into stars.

MongoDB

- Transactional or analytical

- From ERD,
  - Entity set $\longrightarrow$ document class or embed?
  - Avoid lookups via embedding, replicating, pre-computing.
  - ID is synthetic key.

Neo4J

- Transactional or analytical

- From ERD,
  - Entity set → node type
  - Relationship → relationship type
  - Replication & pre-computation less common.
  - ID is synthetic key.